Monash University

Semester Two 2014
Examination Period

Faculty of Engineering

EXAM CODES: **MAE3405**

TITLE OF PAPER: **Flight Vehicle Propulsion**

EXAM DURATION: 3 hours writing time

READING TIME: 10 minutes

THIS PAPER IS FOR STUDENTS STUDYING AT:(tick where applicable)

☐ Berwick	x Clayton	☐ Malaysia	☐ Off Campus Learning	☐ Open Learning
☐ Caulfield	☐ Gippsland	☐ Peninsula	☐ Enhancement Studies	☐ Sth Africa
☐ Parkville	☐ Other (specify)			

During an exam, you must not have in your possession, a book, notes, paper, electronic device/s, calculator, pencil case, mobile phone or other material/item which has not been authorised for the exam or specifically permitted as noted below. Any material or item on your desk, chair or person will be deemed to be in your possession. You are reminded that possession of unauthorised materials, or attempting to cheat or cheating in an exam is a discipline offence under Part 7 of the Monash University (Council) Regulations.

No exam paper or other exam materials are to be removed from the room.

AUTHORISED MATERIALS

OPEN BOOK ☐ **YES** **x NO**

CALCULATORS **x YES** ☐ **NO**
(only calculators with an 'approved for use Faculty of Engineering' or 'Faculty of Science' sticker are permitted)

SPECIFICALLY PERMITTED ITEMS ☐ **YES** **x NO**
if yes, items permitted are:

Candidates must complete this section if required to write answers within this paper

STUDENT ID: __ __ __ __ __ __ __ __ DESK NUMBER: __ __ __ __ __

Note the following: To gain full marks you must: (a) correctly complete all three questions; (b) list assumptions used, provide the necessary diagrams and show all steps in your solutions; and (c) if you use an equation that does not appear on the equation sheet, show how the equation can be derived from those provided, or give a simple derivation of the equation, or show how it can be determined from the relevant thermodynamic cycle.

1. An aircraft is flying steady and level at 5000m with an engine speed of 2500RPM. It is powered by a 6 cylinder, supercharged, intercooled, 4-stroke spark ignition piston engine whose data is tabulated below. The intercooler located after the supercharger reduces the supercharger temperature rise by 50%. Fuel enters the intake after the intercooler. For these conditions, the inlet fuel-air mixture fills the cylinder's swept volume at the intercooler exit conditions. Assuming the engine to operate on the fuel/air Otto cycle and the fuel to be Avgas, determine: **(a)** the engine's power output, and **(b)** the fuel consumption in litres/hour. [23+2=25 Marks]

Quantity	**Value**	**Quantity**	**Value**
Compression ratio	7.5	Combustion efficiency	95%
Bore	0.1m	Air to fuel ratio	14.7
Stroke	0.12m	Cp reactants	1.055kJ/kg-K
Engine mechanical efficiency	88%	Cp products	1.486kJ/kg-K
Supercharger pressure ratio	1.5	Product mole weight	28.0kg/kmol
Supercharger mechanical efficiency	85%	Reactant mole weight	28.9kg/kmol
Supercharger isentropic efficiency	80%	Air: see gas properties table below	

2. A helicopter with a mass of 2 tonnes hovers at an altitude of 1000m. The main rotor with diameter 10m has a blade tip Mach number of Ma=0.5 for this flight condition. The tail rotor with diameter 1.5m is located a distance of 6m from the main rotor shaft. Estimate the power required by the tail rotor to resist the torque generated by main rotor. List any assumptions. [10 Marks]

3. You are required to calculate the thrust for the cruise condition of a new engine design. Known as an aft-fan-ramjet, once at its cruise condition (Ma=1.7, Alt=20,000m) the engine operates as follows. The intake is a supersonic inlet comprised of one oblique shock terminated by a single normal shock. The shock wave angle of the oblique shock is 50^0. A subsonic section directly follows the supersonic intake. After this, similar to a turbofan engine the flow separates into two streams. The hot stream passes directly to a combustor, then on to a turbine (this turbine directly drives the aft-fan), after which it flows to a converging nozzle. The cold stream passes directly to the aft-fan, after which it flows to a converging nozzle. The hot and cold nozzles are separate. Similar to a turbofan engine, the aft-fan is operated at a fixed pressure ratio (FPR) and by-pass for the cruise condition. Operational data is listed below. Assume fuel mass flow is small relative to engine mass flow. [35 Marks]

Quantity	**Value**	**Quantity**	**Value**
FPR	1.8	Combustor pressure drop	4%
Combustor temperature	1900K	Inlet Cp	1.005kJ/kg-K
Isentropic Efficiencies		Inlet γ	1.4
- Fan	85%	Turbine Cp	1.148kJ/kg-K
- Turbine	90%	Turbine γ	1.333
- Subsonic intake section	95%	Exhaust gas constant R	0.287kJ/kg-K
- Propelling nozzles	95%	Cold nozzle area	$0.5m^2$
Combustor efficiency	98%	Fuel	Jet A1
By-pass ratio	3		

Standard Atmosphere

Altitude Km	Temperature K	Pressure Pa	Density kg/m^3	Speed of Sound m/s
0	288.15	101325	1.225	340.29
1	281.65	89874.5	1.111	336.43
2	275.15	79495.2	1.006	332.52
3	268.65	70108.5	0.9091	328.57
4	262.15	61640.2	0.8191	324.57
5	255.65	54019.9	0.7361	320.52
6	249.15	47181.0	0.6597	316.42
7	242.65	41060.7	0.5895	312.27
8	236.15	35599.8	0.5252	308.06
9	229.65	30742.4	0.4663	303.79
10	223.15	26436.2	0.4127	299.46
11	216.65	22632.0	0.3639	295.06
12	216.65	19330.4	0.3108	295.06
13	216.65	16510.4	0.2655	295.06
14	216.65	14101.7	0.2268	295.06
15	216.65	12044.5	0.1937	295.06
16	216.65	10287.4	0.1654	295.06
17	216.65	8786.68	0.1413	295.06
18	216.65	7504.84	0.1207	295.06
19	216.65	6410.00	0.1031	295.06
20	216.65	5474.88	0.088	295.06

Gases Properties

Gas	M_{wt} kg/kmol	R J/kg-K	Cp J/kg-K	Cv J/kg-K	γ	h_f^o MJ/kmole
Air (O2+3.76N2)	29	287.00	1005.0	718.0	1.4	0
Oxygen	32	259.7	921.6	661.9	1.4	0
Carbon dioxide	44	188.92	841.8	652.9	1.289	-393.5
Nitrogen	28	296.80	1041.6	744.8	1.4	0
Water (gas)	18	461.52	1872.3	1410.8	1.327	-241.8

Liquid Fuel Property Data

Fuel	Density kg/m^3	LHV MJ/kg	LHV MJ/litre
Avgas	708.2	43.7	30.9
Jet A1	793.0	43.1	34.2

Planetary Data for the Earth

Quantity	Value
Gravity	$9.81 m/s^2$
Mean Gas Constant	287J/kg-K

Conversion and Reference Data

1Knot = 1.85kph Reference Temperature 298K R_u=8.314 kJ/kmol.K

***Equation Sheet:* state condition numbering used in equations is arbitrary.**

$$S = bc \qquad A = \frac{b}{c} \qquad \mathrm{V_{cal}} = \mathrm{V_E} = \mathrm{V_0}\sqrt{\frac{\rho}{\rho_s}} = \mathrm{V_0}\sqrt{\sigma}$$

$$L = \frac{1}{2}\rho U^2 c_L S \qquad D = \frac{1}{2}\rho U^2 c_D S \qquad c_L = a\alpha \qquad \eta_p = \frac{1}{1+w/V}$$

$$\dot{W} = (IMEP)\dot{V}_s \qquad \dot{V}_s = A_P L_S \frac{N}{2} n \qquad \eta = \frac{\dot{W}}{\dot{Q}_H} \qquad \dot{m} = \rho V A$$

$$TSFC = \frac{\dot{m}_f}{F_N} \qquad BSFC = \frac{\dot{m}_f}{\dot{W}_b} \qquad F_s = \frac{F_N}{\dot{m}_a} \qquad f = \frac{\dot{m}_f}{\dot{m}_a}$$

$$\dot{W}_b = T(w+V) \qquad w = \frac{1}{2}\left[\left(V^2 + \frac{2T}{\rho A_p}\right)^{1/2} - V\right] \qquad \dot{W}_{b,o} = T_o^{\,3/2}/\sqrt{2\rho A_p}$$

$$p = \rho RT \qquad pv = RT \qquad \gamma = C_p/C_v \qquad R = C_p - C_v$$

$$\left(pv^{\gamma}\right)_1 = \left(pv^{\gamma}\right)_2 \qquad \left(Tv^{\gamma-1}\right)_1 = \left(Tv^{\gamma-1}\right)_2$$

$$\sum Q + \sum W = 0 \qquad Q + W = \Delta U \qquad LHV = \left[\left(h_f^o\right)_R - \left(h_f^o\right)_P\right]/M_{wf}$$

$$\left(\frac{p}{\rho} + u + \frac{C^2}{2}\right)_1 = \left(\frac{p}{\rho} + u + \frac{C^2}{2}\right)_2 + Q + W$$

$$\left(\frac{p}{\rho} + u\right) = h \qquad (h_2 - h_1) = C_p(T_2 - T_1) \qquad h = C_p(T - T_R)$$

$$\frac{p_0}{p} = \left(\frac{T_0}{T}\right)^{\frac{\gamma}{\gamma-1}} \qquad \frac{p_{02}}{p_{01}} = \left(\frac{T_{02'}}{T_{01}}\right)^{\frac{\gamma}{\gamma-1}} \qquad \eta_i = \frac{T_{01'} - T_a}{T_{01} - T_a}$$

$$T_{0a} = T_a + \frac{C_a^2}{2c_p} \qquad \frac{p_{01}}{p_a} = \left[1 + \eta_i \frac{C_a^2}{2c_p T_a}\right]^{\frac{\gamma}{\gamma-1}} \qquad TP = \dot{W}_s \eta_p + C_a F_N \qquad \eta_c = \frac{T_{02'} - T_{01}}{T_{02} - T_{01}}$$

$$\eta_t = \frac{T_{03} - T_{04}}{T_{03} - T_{04'}} \qquad \eta_j = \frac{T_{04} - T_5}{T_{04} - T_{5'}} \qquad \frac{p_{04}}{p_c} = \left[1 - \frac{1}{\eta_j}\left(\frac{\gamma-1}{\gamma+1}\right)\right]^{-\left(\frac{\gamma}{\gamma-1}\right)} \qquad \frac{T_c}{T_{04}} = \left(\frac{2}{\gamma+1}\right)$$

$$a = \sqrt{R\gamma T_c} \qquad M = \frac{C}{a} \qquad B = \frac{\dot{m}_c}{\dot{m}_h} \qquad FPR = p_{02}/p_{01} \quad OPR = p_{03}/p_{01}$$

$$C_J = \left[\left(\frac{2\gamma}{\gamma-1}\right)RT_c\left(1-\left(\frac{p_e}{p_c}\right)^{(\gamma-1)/\gamma}\right)\right]^{1/2} \qquad \frac{A}{A^*} = \frac{1}{M}\left(\frac{2+(\gamma-1)M^2}{\gamma+1}\right)^{(\gamma+1)/(2\gamma-2)}$$

$$\Delta C = C_e \ln(M_R) - gt_b \sin\theta \qquad z_b = C_o t_o + C_e t_b\left(\frac{1}{M_R - 1}\ln\left(\frac{1}{M_R}\right)+1\right) - \frac{gt_b^2 \sin\theta}{2}$$

$$z_c = z_b + \frac{r_o C_b^2}{2g_o r_o - C_b^2} \qquad M_R = \frac{M_o}{M_f} \qquad C_{esc} = r_o\sqrt{\left(\frac{2g_o}{r_o+z}\right)} \qquad C_{sat} = r_o\sqrt{\left(\frac{g_o}{r_o+z}\right)}$$

Normal Shock Tables

TABLE E.4 Normal Shock Flow of an Ideal Gas ($k = 1.4$)

Ma_x	Ma_y	$\frac{p_y}{p_x}$	T_y/T_x	$\rho_y/\rho_x = V_x/V_y$	$\frac{p_{0,y}}{p_{0,x}}$	$p_{0,y}/p_x$	Ma_x	Ma_y	$\frac{p_y}{p_x}$	T_y/T_x	$\rho_y/\rho_x = V_x/V_y$	$\frac{p_{0,y}}{p_{0,x}}$	$p_{0,y}/p_x$
1.00	1.00000	1.00000	1.00000	1.00000	1.00000	1.8929	1.49	0.70466	2.4234	1.3136	1.8449	0.93292	3.3756
1.01	0.99013	1.02345	1.00665	1.01669	0.99999	1.9152	1.50	0.70109	2.4583	1.3202	1.8621	0.92978	3.4133
1.02	0.98052	1.04713	1.01325	1.03344	0.99998	1.9379	1.51	0.69758	2.4934	1.3269	1.8792	0.92658	3.4512
1.03	0.97115	1.07105	1.01981	1.05024	0.99997	1.9610	1.52	0.69413	2.5288	1.3336	1.8962	0.92331	3.4894
1.04	0.96202	1.09520	1.02634	1.06709	0.99994	1.9845	1.53	0.69073	2.5644	1.3403	1.9133	0.91999	3.5279
1.05	0.95312	1.1196	1.03284	1.08398	0.99987	2.0083	1.54	0.68739	2.6003	1.3470	1.9303	0.91662	3.5667
1.06	0.94444	1.1442	1.03931	1.10092	0.99976	2.0325	1.55	0.68410	2.6363	1.3538	1.9473	0.91319	3.6058
1.07	0.93598	1.1690	1.04575	1.11790	0.99962	2.0570	1.56	0.68086	2.6725	1.3606	1.9643	0.90970	3.6451
1.08	0.92772	1.1941	1.05217	1.13492	0.99944	2.0819	1.57	0.67768	2.7090	1.3674	1.9812	0.90615	3.6847
1.09	0.91965	1.2194	1.05856	1.15199	0.99921	2.1072	1.58	0.67455	2.7458	1.3742	1.9981	0.90255	3.7245
1.10	0.91177	1.2450	1.06494	1.1691	0.99892	2.1328	1.59	0.67147	2.7828	1.3811	2.0149	0.89889	3.7645
1.11	0.90408	1.2708	1.07130	1.1862	0.99858	2.1588	1.60	0.66844	2.8201	1.3880	2.0317	0.89520	3.8049
1.12	0.89656	1.2968	1.07764	1.2034	0.99820	2.1851	1.61	0.66545	2.8575	1.3949	2.0485	0.89144	3.8456
1.13	0.88922	1.3230	1.08396	1.2206	0.99776	2.2118	1.62	0.66251	2.8951	1.4018	2.0652	0.88764	3.8866
1.14	0.88204	1.3495	1.09027	1.2378	0.99726	2.2388	1.63	0.65962	2.9330	1.4088	2.0820	0.88380	3.9278
1.15	0.87502	1.3762	1.09657	1.2550	0.99669	2.2661	1.64	0.65677	2.9712	1.4152	2.0986	0.87992	3.9693
1.16	0.86816	1.4032	1.10287	1.2723	0.99605	2.2937	1.65	0.65396	3.0096	1.4228	2.1152	0.87598	4.0111
1.17	0.86145	1.4304	1.10916	1.2896	0.99534	2.3217	1.66	0.65119	3.0482	1.4298	2.1318	0.87201	4.0531
1.18	0.85488	1.4578	1.11544	1.3069	0.99455	2.3499	1.67	0.64847	3.0870	1.4369	2.1484	0.86800	4.0954
1.19	0.84846	1.4854	1.12172	1.3243	0.99371	2.3786	1.68	0.64579	3.1261	1.4440	2.1649	0.86396	4.1379
1.20	0.84217	1.5133	1.1280	1.3416	0.99280	2.4075	1.69	0.64315	3.1654	1.4512	2.1813	0.85987	4.1807
1.21	0.83601	1.5414	1.1343	1.3590	0.99180	2.4367	1.70	0.64055	3.2050	1.4583	2.1977	0.85573	4.2238
1.22	0.82998	1.5698	1.1405	1.3764	0.99073	2.4662	1.71	0.63798	3.2448	1.4655	2.2141	0.85155	4.2672
1.23	0.82408	1.5984	1.1468	1.3938	0.98975	2.4961	1.72	0.63545	3.2848	1.4727	2.2304	0.84735	4.3108
1.24	0.81830	1.6272	1.1531	1.4112	0.98835	2.5263	1.73	0.63296	3.3250	1.4800	2.2467	0.84312	4.3547
1.25	0.81264	1.6562	1.1594	1.4286	0.98706	2.5568	1.74	0.63051	3.3655	1.4873	2.2629	0.83886	4.3989
1.26	0.80709	1.6855	1.1657	1.4460	0.98568	2.5876	1.75	0.62809	3.4062	1.4946	2.2791	0.83456	4.4433
1.27	0.80165	1.7150	1.1720	1.4634	0.98422	2.6187	1.76	0.62570	3.4472	1.5019	2.2952	0.83024	4.4880
1.28	0.79631	1.7448	1.1782	1.4808	0.98268	2.6500	1.77	0.62335	3.4884	1.5093	2.3113	0.82589	4.5330
1.29	0.79108	1.7748	1.1846	1.4983	0.98106	2.6816	1.78	0.62104	3.5298	1.5167	2.3273	0.82152	4.5783
1.30	0.78596	1.8050	1.1909	1.5157	0.97935	2.7135	1.79	0.61875	3.5714	1.5241	2.3433	0.81711	4.6238
1.31	0.78093	1.8354	1.1972	1.5331	0.97758	2.7457	1.80	0.61650	3.6133	1.5316	2.3592	0.81268	4.6695
1.32	0.77600	1.8661	1.2035	1.5505	0.97574	2.7783	1.81	0.61428	3.6554	1.5391	2.3751	0.80823	4.7155
1.33	0.77116	1.8970	1.2099	1.5680	0.97382	2.8112	1.82	0.61209	3.6978	1.5466	2.3909	0.80376	4.7618
1.34	0.76641	1.9282	1.2162	1.5854	0.97181	2.8444	1.83	0.60993	3.7404	1.5542	2.4067	0.79926	4.8083
1.35	0.76175	1.9596	1.2226	1.6028	0.96972	2.8778	1.84	0.60780	3.7832	1.5617	2.4224	0.79474	4.8551
1.36	0.75718	1.9912	1.2290	1.6202	0.96756	2.9115	1.85	0.60570	3.8262	1.5694	2.4381	0.79021	4.9022
1.37	0.75269	2.0230	1.2354	1.6376	0.96534	2.9455	1.86	0.60363	3.8695	1.5770	2.4537	0.78567	4.9498
1.38	0.74828	2.0551	1.2418	1.6550	0.96304	2.9798	1.87	0.60159	3.9130	1.5847	2.4693	0.78112	4.9974
1.39	0.74396	2.0874	1.2482	1.6723	0.96065	3.0144	1.88	0.59957	3.9568	1.5924	2.4848	0.77656	5.0453
1.40	0.73971	2.1200	1.2547	1.6896	0.95819	3.0493	1.89	0.59758	4.0008	1.6001	2.5003	0.77197	5.0934
1.41	0.73554	2.1528	1.2612	1.7070	0.95566	3.0844	1.90	0.59562	4.0450	1.6079	2.5157	0.76735	5.1417
1.42	0.73144	2.1858	1.2676	1.7243	0.95306	3.1198	1.91	0.59368	4.0894	1.6157	2.5310	0.76273	5.1904
1.43	0.72741	2.2190	1.2742	1.7416	0.95039	3.1555	1.92	0.59177	4.1341	1.6236	2.5463	0.75812	5.2394
1.44	0.72345	2.2525	1.2807	1.7589	0.94765	3.1915	1.93	0.58988	4.1790	1.6314	2.5615	0.75347	5.2886
1.45	0.71956	2.2862	1.2872	1.7761	0.94483	3.2278	1.94	0.58802	4.2242	1.6394	2.5767	0.74883	5.3381
1.46	0.71574	2.3202	1.2938	1.7934	0.94196	3.2643	1.95	0.58618	4.2696	1.6473	2.5919	0.74418	5.3878
1.47	0.71198	2.3544	1.3004	1.8106	0.93901	3.3011	1.96	0.58437	4.3152	1.6553	2.6070	0.73954	5.4378
1.48	0.70829	2.3888	1.3070	1.8278	0.93600	3.3382	1.97	0.58258	4.3610	1.6633	2.6220	0.73487	5.4880

Normal Shock Tables Continued

TABLE E.4 (continued)

Ma_x	Ma_y	$\frac{p_y}{p_x}$	T_y/T_x	$\rho_y/\rho_x = V_x/V_y$	$\frac{p_{0,y}}{p_{0,x}}$	$p_{0,y}/p_x$	Ma_x	Ma_y	$\frac{p_y}{p_x}$	T_y/T_x	$\rho_y/\rho_x = V_x/V_y$	$\frac{p_{0,y}}{p_{0,x}}$	$p_{0,y}/p_x$
1.98	0.58081	4.4071	1.6713	2.6369	0.73021	5.5385	2.55	0.50831	7.4196	2.1875	3.3918	0.47927	8.8505
1.99	0.57907	4.4534	1.6794	2.6518	0.72554	5.5894	2.56	0.50740	7.4792	2.1976	3.4034	0.47540	8.9162
2.00	0.57735	4.5000	1.6875	2.6666	0.72088	5.6405	2.57	0.50651	7.5391	2.2077	3.4149	0.47155	8.9821
2.01	0.57565	4.5468	1.6956	2.6814	0.71619	5.6918	2.58	0.50562	7.5992	2.2179	3.4263	0.46772	9.0482
2.02	0.57397	4.5938	1.7038	2.6962	0.71152	5.7434	2.59	0.50474	7.6595	2.2281	3.4376	0.46391	9.1146
2.03	0.57231	4.6411	1.7120	2.7109	0.70686	5.7952	2.60	0.50387	7.7200	2.2383	3.4489	0.46012	9.1813
2.04	0.57068	4.6886	1.7203	2.7255	0.70218	5.8473	2.61	0.50301	7.7808	2.2486	3.4602	0.45636	9.2481
2.05	0.56907	4.7363	1.7286	2.7400	0.69752	5.8997	2.62	0.50216	7.8418	2.2589	3.4714	0.45262	9.3154
2.06	0.56747	4.7842	1.7369	2.7545	0.69284	5.9523	2.63	0.50132	7.9030	2.2693	3.4825	0.44891	9.3829
2.07	0.56589	4.8324	1.7452	2.7690	0.68817	6.0052	2.64	0.50048	7.9645	2.2797	3.4936	0.44522	9.4507
2.08	0.56433	4.8808	1.7536	2.7834	0.68351	6.0584	2.65	0.49965	8.0262	2.2901	3.5047	0.44155	9.5187
2.09	0.56280	4.9295	1.7620	2.7977	0.67886	6.1118	2.66	0.49883	8.0882	2.3006	3.5157	0.43791	9.5869
2.10	0.56128	4.9784	1.7704	2.8119	0.67422	6.1655	2.67	0.49802	8.1504	2.3111	3.5266	0.43429	9.6553
2.11	0.55987	5.0275	1.7789	2.8216	0.66957	6.2194	2.68	0.49722	8.2128	2.3217	3.5374	0.43070	9.7241
2.12	0.55830	5.0768	1.7874	2.8402	0.66492	6.2736	2.69	0.49642	8.2754	2.3323	3.5482	0.42713	9.7932
2.13	0.55683	5.1264	1.7960	2.8543	0.66029	6.3280	2.70	0.49563	8.3383	2.3429	3.5590	0.42359	9.8625
2.14	0.55538	5.1762	1.8046	2.8683	0.65567	6.3827	2.71	0.49485	8.4014	2.3536	3.5697	0.42007	9.9320
2.15	0.55395	5.2262	1.8132	2.8823	0.65105	6.4377	2.72	0.49408	8.4648	2.3643	3.5803	0.41657	10.002
2.16	0.55254	5.2765	1.8219	2.8962	0.64644	6.4929	2.73	0.49332	8.5284	2.3750	3.5909	0.41310	10.072
2.17	0.55114	5.3270	1.8306	2.9100	0.64185	6.5484	2.74	0.49256	8.5922	2.3858	3.6014	0.40965	10.142
2.18	0.54976	5.3778	1.8393	2.9238	0.63728	6.6042	2.75	0.49181	8.6562	2.3966	3.6119	0.40622	10.212
2.19	0.54841	5.4288	1.8481	2.9376	0.63270	6.6602	2.76	0.49107	8.7205	2.4074	3.6224	0.40282	10.283
2.20	0.54706	5.4800	1.8569	2.9512	0.62812	6.7163	2.77	0.49033	8.7850	2.4183	3.6328	0.39945	10.354
2.21	0.54572	5.5314	1.8657	2.9648	0.62358	6.7730	2.78	0.48960	8.8497	2.4292	3.6431	0.39610	10.426
2.22	0.54440	5.5831	1.8746	2.9783	0.61905	6.8299	2.79	0.48888	8.9147	2.4402	3.6533	0.39276	10.498
2.23	0.54310	5.6350	1.8835	2.9918	0.61453	6.8869	2.80	0.48817	8.9800	2.4512	3.6635	0.38946	10,569
2.24	0.54182	5.6872	1.8924	3.0052	0.61002	6.9442	2.81	0.48746	9.0454	2.4622	3.6737	0.38618	10.641
2.25	0.54055	5.7396	1.9014	3.0186	0.60554	7.0018	2.82	0.48676	9.1111	2.4733	3.6838	0.38293	10.714
2.26	0.53929	5.7922	1.9104	3.0319	0.60106	7.0597	2.83	0.48607	9.1770	2.4844	3.6939	0.37970	10.787
2.27	0.53805	5.8451	1.9194	3.0452	0.59659	7.1178	2.84	0.48538	9.2432	2.4955	3.7039	0.37649	10.860
2.28	0.53683	5.8982	1.9285	3.0584	0.59214	7.1762	2.85	0.48470	9.3096	2.5067	3.7139	0.37330	10.933
2.29	0.53561	5.9515	1.9376	3.0715	0.58772	7.2348	2.86	0.48402	9.3762	2.5179	3.7238	0.37013	11.006
2.30	0.53441	6.0050	1.9468	3.0846	0.58331	7.2937	2.87	0.48334	9.4431	2.5292	3.7336	0.36700	11.080
2.31	0.53322	6.0588	1.9560	3.0976	0.57891	7.3529	2.88	0.48268	9.5102	2.5405	3.7434	0.36389	11.154
2.32	0.53205	6.1128	1.9652	3.1105	0.57452	7.4123	2.89	0.48203	9.5775	2.5518	3.7532	0.36080	11.228
2.33	0.53089	6.1670	1.9745	3.1234	0.57015	7.4720	2.90	0.48138	9.6450	2.5632	3.7629	0.35773	11.302
2.34	0.52974	6.2215	1.9838	3.1362	0.56580	7.5319	2.91	0.48074	9.7127	2.5746	3.7725	0.35469	11.377
2.35	0.52861	6.2762	1.9931	3.1490	0.56148	7.5920	2.92	0.48010	9.7808	2.5860	3.7821	0.35167	11.452
2.36	0.52749	6.3312	2.0025	3.1617	0.55717	7.6524	2.93	0.47946	9.8491	2.5975	3.7917	0.34867	11.527
2.37	0.52638	6.3864	2.0119	3.1743	0.55288	7.7131	2.94	0.47883	9.9176	2.6090	3.8012	0.34570	11.603
2.38	0.52528	6.4418	2.0213	3.1869	0.54862	7.7741	2.95	0.47821	9.986	2.6206	3.8106	0.34275	11.679
2.39	0.52419	6.4974	2.0308	3.1994	0.54438	7.8354	2.96	0.47760	10.055	2.6322	3.8200	0.33982	11.755
2.40	0.52312	6.5533	2.0403	3.2119	0.54015	7.8969	2.97	0.47699	10.124	2.6438	3.8294	0.33692	11.831
2.41	0.52206	6.6094	2.0499	3.2243	0.53594	7.9587	2.98	0.47638	10.194	2.6555	3.8387	0.33404	11.907
2.42	0.52100	6.6658	2.0595	3.2366	0.53175	8.0207	2.99	0.47578	10.263	2.6672	3.8479	0.33118	11.984
2.43	0.51996	6.7224	2.0691	3.2489	0.52758	8.0830	3.00	0.47519	10.333	2.6790	3.8571	0.32834	12.061
2.44	0.51894	6.7792	2.0788	3.2611	0.52344	8.1455	3.50	0.45115	14.125	3.3150	4.2608	0.21295	16.242
2.45	0.51792	6.8362	2.0885	3.2733	0.51932	8.2083	4.00	0.43496	18.500	4.0469	4.5714	0.13876	21.068
2.46	0.51691	6.8935	2.0982	3.2854	0.51521	8.2714	4.50	0.42355	23.458	4.8751	4.8119	0.09170	26.539
2.47	0.51592	6.9510	2.1080	3.2975	0.51112	8.3347	5.00	0.41523	29.000	5.8000	5.0000	0.06172	32.654
2.48	0.51493	7.0088	2.1178	3.3095	0.50706	8.3983	6.00	0.40416	41.833	7.941	5.2683	0.02965	46.815
2.49	0.51395	7.0668	2.1276	3.3214	0.50303	8.4622	7.00	0.39736	57.000	10.469	5.4444	0.01535	63.552
2.50	0.51299	7.1250	2.1375	3.3333	0.49902	8.5262	8.00	0.39289	74.500	13.387	5.5652	0.00849	82.865
2.51	0.51204	7.1834	2.1474	3.3451	0.49502	8.5904	9.00	0.38980	94.333	16.693	5.6512	0.00496	104.753
2.52	0.51109	7.2421	2.1574	3.3569	0.49104	8.6549	10.00	0.38757	116.50	20.388	5.7143	0.00304	129.217
2.53	0.51015	7.3010	2.1674	3.3686	0.48709	8.7198	∞	0.37796	∞	∞	6.000	0	∞
2.54	0.50923	7.3602	2.1774	3.3802	0.48317	8.7850							

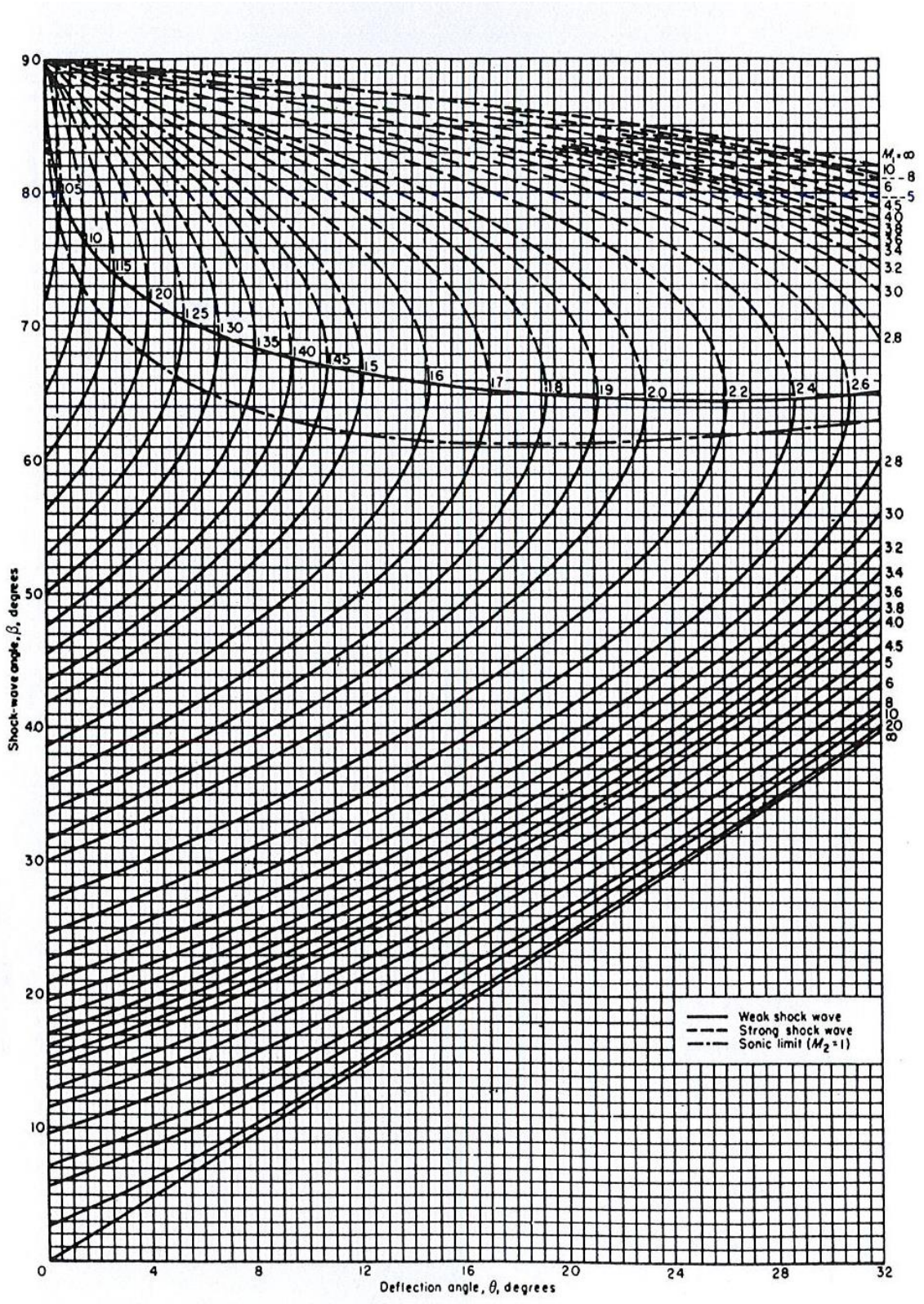


FIGURE 9.7
Oblique shock properties: $\gamma = 1.4$. The θ-β-M diagram. (*From NACA Report 1135, Ames Research Staff, "Equations, Tables and Charts for Compressible Flow," 1953.*)